

2 Fundamentals of a Computer System

18 February 2022

We can classify the computers

1. Based on size

a. Super Computers



2. Main Frame Computers



c. Mini Computers



d. Micro Computer

Laptops and Desktops are classified as Microcomputers. We can easily remember them as Microsoft means **Micro** Computer **Software**. As these devices mostly use Microsoft software's we can remember it as Micro Computers in the examination.



2. Based on technology

a. Analogue/Analog Computers

Analogue computer is a type of computers that uses the continuously changeable aspects of physical phenomena such as electrical, mechanical or hydraulic qualities to model the problem being solved. Though once popular they lost the popularity due to the advancement of Digital Computers.

b. Digital Computers

Digital computers is a Computer Machine which is both electronic computer and digital computer. Almost all new computers digital computers.

c. Hybrid Computers

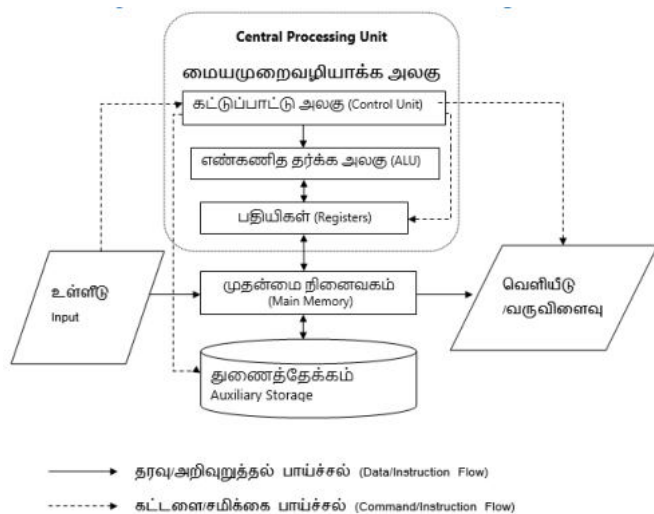
Hybrid computers exhibit the features of Digital and Analogue computers.

3. CPU

CPU is the brain of the computers. Speed of the computers (CPU) measured is GHz (Giga Hertz) or Mhz (Mega Hertz). MIPS (Millions of Instructions Per Second) used previously and not used now.

4. Below architecture also known as Von Numan's Architecture as Numan proposed Stored Program Concept initially.

CPU consists of Arithmetic and Logic Unit, Control Unit and Registers. L1 (Level 1) cache also exists inside the CPU. Cache was created to speed up the execution but it's speed is less than that of the Register. Control Unit gets the instructions and decode the instructions.



5. Components of Computers

- Physical Components - The components that we can touch like Keyboard and mouse.
- Software - We cannot touch.
- Firmware - They help the computer to start. Generally stored in BIOS (Basic Input Output System).
- Liveware - Computer Users are referred to as Liveware.

6. Physical Components of Computers

a. Input Devices

i. Keyboard

Picture below shows with the finger positions (Numeric pad is not shown)



2. Mouse 🖱️

These days most of the mouse's are optical mouse's they track the movements using InfraRed light.



iii. Touchpad - Mostly used in laptop also exists in early smartphones.



iv. Trackball 🖱️ - Looks like an inverted mouse (These days mouse don't have balls inside but early mice do have this part).



v. Joystick 🕹️



vi. Microphone 🎤



vii. Scanner



viii. Bar Code Reader



ix. Input Components that do have the ability to point is known as the pointing devices

e,g Mouse, Joystick and Trackball.

b. Output Devices

i. Monitor ☐



b. Printer/Print Device ☐



iii. Speakers ☐ (Picture below shows with AC input but most of the modern speakers have 5V DC input)



iv. Multimedia Projectors ☐

1. Overhead Projectors

Overhead Projectors uses transparent sheet s to project the content.



ii. Slide Project uses slide to project



3. Multimedia projectors - Multimedia projectors used to project the content from Computer/Laptop. With the advancement of multimedia projectors overhead and slide projectors lost its popularity.



c. **Input and Output Devices**

Touch screen serves as an Input and Output Device. Almost all the smartphones do have the touch screens and some of the expensive laptops do have the touch screen. Smartboards used in the Schools do the touch facility.



7. **Computer Memory**

a. **Primary Memory**

- i. **Registers** - Exists inside the CPU. It's the fastest memory. Smallest capacity. It can only keep the data during the processing when the computer is switch off it will loose the content so it's known as a volatile memory.
- ii. **Cache** - L1 (Level 1) cache exists inside the CPU other types of cache memory exists outside the CPU. It's fast but not fast as register. It can only keep the data during the processing when the computer is switch off it will lose the content so it's known as a volatile memory.
- iii. **ROM** - stands for Read Only Memory. It consists of firmware used to start the computer. Content cannot be changed. Older motherboards BIOS (Basic Input Output System) can be seen easily and with newer mother boards it's embedded inside the motherboard mostly cannot see. It will not lose the content when the computer is switched off so it's also known as nonvolatile memory.
- iv. **RAM**- stands for Random Access Memory. It keeps the content during the computer is working and it will lose the content when the computer is switched off. So its known as volatile memory.
- v. Note only for the OL examination we can generally refer the Primary memory as Volatile memory. This will not accepted in AL ICT.

b. **Secondary Memory**

i. **Magnetic Media**

1. **Hardisks** ≡ Hardisk capacity varies from GB to TB.
2. **Floppy disks**  though once popular now almost obsolete due to it's small storage (1.44-2.88MB). It's been replaced with pen/usb drive. Though doesn't exists today but most of the application save symbols shows floppy disks icon.
3. **Magnetic Tapes**  As the access method is sequential mostly used to backup purposes. This is device is having the highest storage capacity.

ii. **Optical Media**

1. CD (Compact Disks)
2. DVD (Digital Versatile Disk)
3. BlueRay Disk

iii. **Electronic Media**

1. **SSD** - Solid State Drives
2. Pendrive/Flash Drive
3. Micro SD cards

8. **Ports used in Computers**

- a. **PS/2 Ports** - Introduced by IBM to connect Keyboard and Mouse only.



2. **Parallel Ports**- Can only be used to connect printers.



- c. **VGA Ports** - Video Graphics Array. Can be used to connect Monitor, TV, Multimedia Projectors.



- d. **HDMI** (High Definition Multimedia Interface) ports. As multimedia means Video and Audio this can transmit Video and Audio through this port.



- e. **Serial /Com ports** -



- f. **USB port** - Universal Serial Bus is becoming more popular and try to replace all other ports. Picture below shows two USB ports.

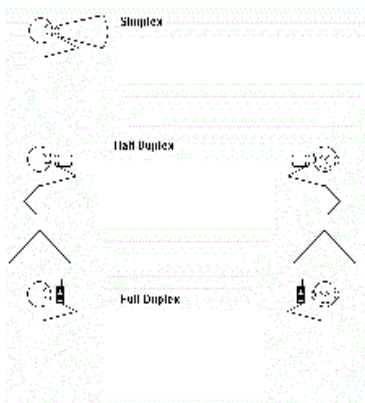


9. **Data Bus** in the motherboard helps data communications. You can remember this by looking at the Bus on the road as it transports the passengers here in the Computers you can think as transporting data.
10. **PDA** - Personal Digital Assistant was once popular devices that fit inside the palm. Due to the advancements of Smartphone, they replaced the PDA.



11. **Data Communication**

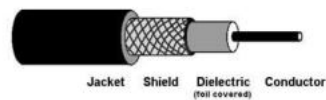
- a. **Simplex** - In TV, Radio Broadcasting simplex communication takes place. As they broadcast we only listen.
- b. **Half Duplex** - In Walkie Talkie During the communication when one party finish communicating the party will say over then the other party communicates.
- c. **Full Duplex** - Telephone can be considered as full duplex as both parties can communicate at the same time.



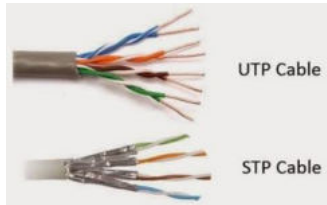
12. **Data Communication Media**

- a. **Guided/Wired Media**
- i. **Coaxial Cable** - Coaxial cables are similar to TV antennas .

Coaxial Cable Construction



2. Twisted Pair - During the early days of communication electricity was passed through copper cable generated electromagnetic which interfered with the communication and hence noise occurred. In order to reduce the noise the cables were twisted so that as one cable's electromagnetic interference cancels the other. They are cheap. They are limited to 100m cable length. If you want to extend the range then need to connect via a repeater. They are classified as two.
 - a. UTP - Unshielded Twisted Pair
 - b. STP - Shielded Twisted Pair



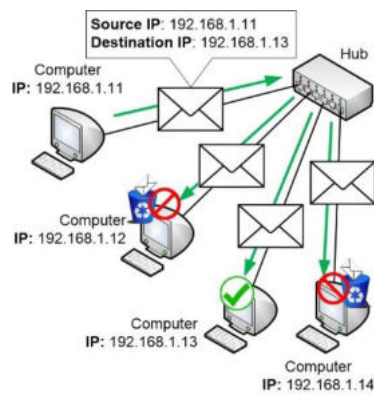
3. Fiber Optics - These cables use total internal reflection property of light so that they can transmit data to very long distance with almost no loss of data at the very high speed. These are the fastest media. As they are not affected by electromagnetic fields they are most preferred but they are expensive than UTP/STP cables. Undersea cables connecting countries are using this technology.



- b. Unguided/Wireless Media
 - i. Radio waves - Bluetooth and Wi-Fi are considered as wireless.
 - ii. Micro waves - Mostly used to communicate between telecom tower and satellite. They need to be in line of sight (needs to be in the same direction) in order to communicate. Microwaves can penetrate the clouds they are much preferred as they can work almost any weather condition.
 - iii. Infrared - TV remotes, Multimedia remotes use Infra red waves to communicate. They need to be in the line of sight in order to communicate.
- m. **Advantages of Networking**
 - i. Sharing of resources - Networks can be utilized to share resources like printer as printer can share across the network.
 - ii. Reduce the data redundancy - Files can be copied to a central location and the computers connected to the network can access the files instead of each computer copying the files individually.
 - iii. Reduce the data communication cost - One internet connection can be used by all the computers connected through the network reduce the data communication cost.
- n. **Disadvantages of Networking**
 - i. Computer virus may spread through network.
 - ii. Technically Qualified staff may be required to maintain the network.
- o. **Equipments used in Networking**
 - i. NIC - Network Interface Card Wired media generally uses RJ45 (Registered Jack 45) for network connection. They have 8 pins. Internal Network Card shown below.



- b. Switch/Hub - Switches are preferred than hub due to the fact they are intelligent as they send the data through the required ports. Hub is multiport repeater that they send the packets to all other ports.



16. Network topologies

- a. **Bus Topology** - In bus topology devices are connected through a cable backbone.



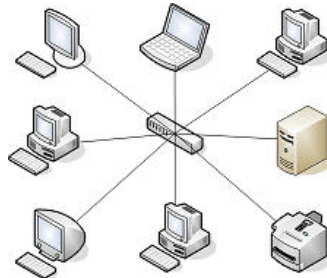
1. Advantages

- a. Requires less cable compared to star topology so it requires less cable.
- b. If single computer or device malfunctioned will not affect the rest of the network.

2. Disadvantages

- a. If the backbone failure occurs then the whole network will be affected.
- b. If any problems happens in bus topology it's difficult to find where the fault is.

2. **Star Topology** - Each device is connected to central switch. This is the most widely used network topology today.



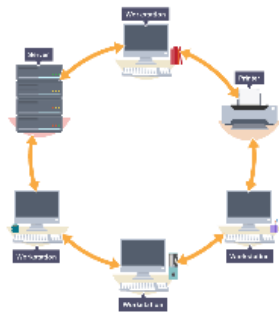
i. Advantages

- i. Compared to Bus topology this one is more efficient as they need to send only to the required receptionist.
- ii. Easily can add or remove devices to the network.
- iii. As each device is connected through a separate cable this offers good speed for the connection.
- iv. Easy to troubleshoot (easy to find the problem and fix the problem).

ii. Disadvantages

- i. Since each device is connected via a single cable cost of the connection is high.
- ii. If the switch/hub fails total network failure occurs.
- iii. How many devices can be connected to the network depends on how many ports switch have.

- iii. **Ring Topology** - Each device is connected to two other devices in this network. Token is passed through the network and when there is an empty token they can transmit the data. This was invented by IBM though cooperation adapted but due to the advancements of Ethernet they lost its popularity. Here each node is connected with exactly two other computers.



- i. Advantages
 - i. Only small cable is enough to connect to the network.
 - ii. Since only they can start communication when an empty token arrives collision of data packets are avoided. (CSMA/CA - Carrier Sense Multiple Access with Collision Avoidance).
 - iii. Though the load is increased when connecting more device but it can manage better than Bus topology.
- ii. Disadvantages
 - i. If a single device in the network fails affects the whole network.
 - ii. Difficult to troubleshoot (Difficult to find the faults and repair the connection).
 - iii. This network uses expensive components.
- iv. **Mesh Topology** - each device is connected each other device is known as the mesh topology.
 - i. Advantages
 - i. Since each device is connected to each other device there won't be any network failure.
 - ii. We can send multiple messages at the same time across the devices connected to this network.
 - iii. Easy to troubleshoot as each device is connected a single cable to each other computer.
 - ii. Disadvantages
 - i. Expensive as the total number of cables required is $n(n-1)/2$ also wires looks clumsy. Also needs additional I/O (input output) ports increase the cost further.
 - ii. Difficult to install and troubleshoot.
- xvii. Based on geographical spread we can classify the networks as. For OL exam point of view LAN, MAN, WAN should be enough.

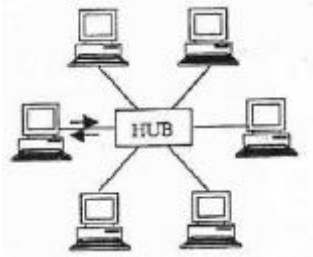
	PAN	LAN	MAN	WAN
Abbreviation	Personal Area Network	Local Area Network	Metropolitan Area Network	Wide Area Network
Geographical Spread	Generally a room size	Buildings or Building Complexes	Connected between cities	Connected across the countries
Technology	Bluetooth, Wi-Fi, USB	Ethernet cables (with RJ45 pins), Wi-Fi	Generally connected via Telecom service provide like SLT	Fiber Optics
Examples	Laptop connected with a Bluetooth speaker	School Computer Lab	Banks	Internet

18. Unitwise MCQ questions can be done via <http://ictmcq.home.blog>

MCQ

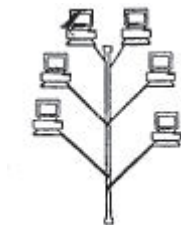
1. The most general term used to describe the physical components of a computer is OL 2007MCQ03
(1) Software (2) Hardware (3) Firmware (4) Liveware
2. The part of a computer that processes instructions, performs calculations, makes logical decisions and manages the flow of data through the computer system is known as OL 2007MCQ04
(1) Hard Disk (2) RAM (3) Data Bus (4) CPU

3. Consider the following statements OL 2007MCQ05.
 A - Keyboard is an input device
 B - Monitor is an output device
 C - Compact disk is a storage device
 Which of the above is/are true?
 (1) A only (2) B only (3) A and C only (4) All A, B and C
4. Under which of the following categories is a laptop computer classified? OL 2007MCQ06
 (1) Microcomputer (2) Minicomputer (3) Mainframe (4) Supercomputer
5. The clock rate of a CPU of a modern Personal Computer (PC) is usually expressed in OL 2007MCQ08
 (1) GHz (gigahertz) (2) Mbps (megabits per second)
 (3) GB (gigabytes). (4) IPS (Instruction Per Second).
6. Which of the following statements is **incorrect** regarding the computer memory? OL 2007MCQ09
 (1) Memory refers to the primary memory or internal memory of the computer
 (2) It holds data & program instruction for processing.
 (3) If the electrical power is cut off then the stored data & instruction will be lost.
 (4) It is the place where arithmetic and logic operations are done.
7. A computer network which is limited to a small area is known as a OL 2007MCQ10
 (1) LAN (2) WAN (3) VAN (4) MAN.
8. What is the topology of the computer network shown in the following diagram? OL2007MCQ11



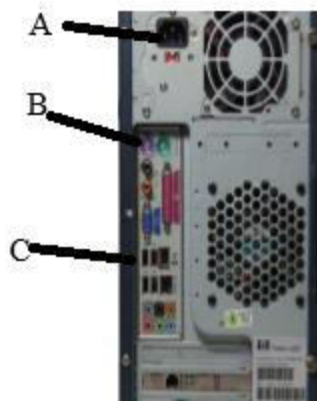
- (1) Bus
 - (2) Star
 - (3) Ring
 - (4) Tree
9. Which of the following problems have been solved by Computer Networks in an office environment? OL2007MCQ30
 A - Difficulty in sharing of resources.
 B - Data redundancy
 C - Cost of communication
 (1) A and C only (2) A and B only (3) B and C only (4) All A, B, and C
10. Which of the following is **not** a port in computer system? GIT 2007MCQ36
 1)COM 1 2)Joystick 3)MIC in 4)PS/2
11. Select the most correct statement? GIT 2007MCQ38
 1)Laser printer is an output device 2)Speaker is an input/output device
 3)Mouse is an output device 4)Keyboard is an output device
12.is a very small computer that can be held in the palm of a person. OL2008MCQ02
 (1) PC (2) Laptop (3) PDA (4) Minicomputer
13. Which one of the following is **not** a pointing device? OL2008MCQ03
 (1) Light pen (2) Joystick (3) Trackball (4) Barcode reader
14. Television broadcast can be considered as an example of..... communication. OL2008MCQ06
 (1) simplex (2) half duplex (3) full duplex (4) point to point

15. What is the topology of the computer network shown in Figure? OL2008MCQ07



- (1) Bus
(2) Star
(3) Ring.
(4) Tree
16. In computer applications, the acronym CAD generally stands for: OL2008MCQ35
(1) Computer Aided Drawing (2) Computer Aided Design
(3) Computer And Data (4) Customized Application Design
17. Local area network (LAN) can be protected from unauthorized access through the Internet using the following GIT2008MCQ35
1)Virus Scanner 2)Disk Defragmenter 3)Firewall 4)Trojan Horse
18. Which port **cannot** be used to connect a printer to the computer? GIT2008MCQ37
1)PS2 2)Serial 3)USB 4)Parallel
19. Which of the following is the most important component to surf the Internet using a home computer? GIT2008MCQ38
1)Network Printer 2)Virus Guard 3)Web Brower 4)Firewall
20. The following is **not** an essential basic components of a computer network? GIT2008MCQ39
1)Data communication peripheral devices such as modems, routers
2)Two or more computers
3)A data communication media such as copper transmission cables
4)Software for email communication
21. Which row of the given table matches the two columns, Hardisk and Scanner correctly? OL2009MCQ03
- | Row | Hard disk | Scanner |
|-----|-----------|---------|
| A | Input | Output |
| B | Storage | Input |
| C | Input | Storage |
| D | Output | Input |
- 1)A 2)B 3)C 4)D
22. The management of a certain cafeteria expects to offer a better service to their customers. As the first step they want to find out how long it has to be kept opened to serve food to all customers during lunch time. They hope to input the following data into a computer with relevant software in this regard. OL2009MCQ05
(A) The number of customers using the cafeteria during lunch time.
(B) The number of cahiers
(C) The average time that a customer spends waiting in the queue.
Which of the above should be input to the computer?
1) (A) and (B) 2)(A) and (C) 3)(B) and (C) 4)All (A), (B) and (C)
23. Travelling of two trains in opposite directions through a single tunnel in a railway can be used as an example to explain: OL2009MCQ06
(1) simplex communication of data through a data channel
(2) half duplex communication of data through a data channel
(3) full duplex communication of data through a data channel
(4) point to point communication of data through a data channel
24. Which one of the following is **incorrect** regarding Local Area Network (LANs)?OL2009MCQ07
1)Geographically distributed in a small area
2)Generally owned by a single person or an organization
3)Consist of many shared data communication medium with various interfaces
4)Generally consists of topologies: Bus, Star or Ring

25. A computer responsible for storing and managing data centrally is a computer network is called a/an OL 2009MCQ35
1)file server 2)data server 3)web server 4)application server
26. Which of the following is **not** used to enter data into a computer? GIT2009MCQ34
1)Wireless Mouse 2)Scanner 3)Laser Printer 4)Touch Screen
27. Which of the following is the least important component to browse webpages on the Internet? GIT 2009MCQ35
1)Printer 2)Web Browser 3)Data Communication Device 4)Computer
28. Which of the following is **false** regarding the input device? GIT2009MCQ37
1)A track ball is similar to a mouse device upside down.
2)Trackpads are commonly found in laptop computers.
3)A mouse is an input device that translate its movement on the desktop into analog form.
4)A graphics tablet can be used to trace over technical drawings.
29. Which of the following is **not** basic component of Local Area Network (LAN)? GIT2009MCQ38
1)A webserver 2)Transmission Media 3)Communication Properties 4)Two or more Computers
30. Consider the following computer devices: OL2010MCQ03
A - Keyboard B - Scanner C - Printer D - Speaker
Which of the above could be used as input derives of a computer?
1)A and B only 2)A and C only 3)C and D only 4)All A, B, C and D
31. Consider the following statements: OL2010MCQ04
A - Random Access Memory (RAM) is a volatile memory.
B - Hard disk is an auxiliary storage device.
C - A floppy disk can store more data than a compact disk (CD)
Which of the above is/are correct?
1)A only 2)C only 3)A and B only 4)B and C only
32. Identify A, B and C from figure OL2010MCQ05



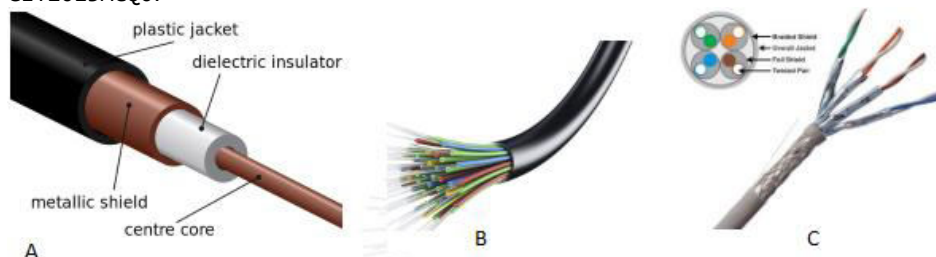
- 1)A = power socket, B= PS/2 port, C= USB port
 - 2)A = power socket, B = serial port, C = PS/2 port
 - 3)A = power socket, B = parallel port, C = Serial Port
 - 4)A = parallel port, B = serial port. C=USB port
33. The Central Processing Unit(CPU) consists of GIT2010MCQ03
1)Arithmetic and Logic Unit (ALU) and Control Unit (CU).
2)Arithmetic and Logic Unit (ALU) and Input Unit.
3)Input Unit and Output Unit.
4)Control Unit (CU) and Input Unit.
 34. The most suitable device for Data Entry Operator who does data entry regularly is a GIT2010MCQ05
1)Palmtop Computer 2)Laptop Computer 3)Desktop Computer 4)Touch Screen

35. Which of the following types of networks can be used to connect a number of branches of a bank scattered around the country. GIT2010MCQ10
a - Local Area Network b - Wide Area Network c - Metropolitan Area Network
1)a only 2)b only 3)c only 4)b and C only
36. Which of the following component is most useful in connecting a computer to a Local Area Network? GIT2010MCQ11
1)Router 2)Monitor 3)Network Interface Card 4)Modem
37. Which of the following devices could be connected to a Local Area Network? GIT2010MCQ12
a - Printer b - Scanner c - Server
1)a only 2)c only 3)a and c only 4)All a, b, c
38. Which of the following statements is/are relating to media used for data communication? GIT 2010MCQ15
a - Fibre optics cables provide faster transfer rates than copper cables.
b - Copper cables provide faster transfer rates than fibre optic cables.
c - Copper cables are expensive than the fibre optics cables.
1)a only 2)a and c only 3)b only 4)b and c only
39. A manager of a certain company would like to protect his computer from unauthorized access by others through Internet. Which of the following is the most suitable solution for his requirement? GIT2010MCQ38
1)Installing an antivirus software 2)Upgrading the version of the operating system
3)Installing firewall 4)Avoiding the use of pirated software
40. A student prepares a report and saves into a permanent storage device (device 1) inside the computer. He then copies the report to a portable storage device (device 2) to be given to a friend. Which of the following is most appropriate to represent device 1 and device 2 respectively?OL2011MCQ02
1)Main memory and flash Memory 2)Hard disk and Compact Disk (CD)
3)Communication device only 4)Input/output device
41. A person takes photograph using a digital camera. Subsequently he shows photograph to one of his friends using the display of the camera. In this scenario, the camera serves as a/an OL2011MCQ03
1)Input device only 2)Output device only
3)Communication device only 4)Input/Output device only
42. Which of the following combinations consist of an input device, an output device and a storage device respectively GIT2011MCQ03
1)Keyboard, Scanner, Mouse 2)Scanner, Mouse, Keyboard
3)Compact Disk Drive, Keyboard, Flash Memory 4)Keyboard, Speaker, Compact Disk Drive
43. A salesman who usually travels by motorcycle needs to use (carry) a computer to record the sales orders he takes from the shops he visits. Which of the following is/are suitable for this purpose? GIT2011MCQ07
(A) - Desktop Computer (B) - Palmtop Computer (C) - Laptop Computer
1)A only 2)B only 3)Both A and B 4)Both B and C
44. The central processing Unit (CPU) of a computer consists of GIT2011MCQ09
1)Control Unit and Arithmetic and Logic Unit (ALU) 2)Control Unit and Output Unit
3)Main Memory and Auxiliary Storage 4)Main Memory and Output Unit
45. Which of the following are network communication device? GIT2011MCQ17
1)Network Interface Card, Local Area Network (LAN) 2)Router, Wide Area Network (WAN)
3)Switch, Router 4)Network Interface Card, Wide Area Network (WAN)
46. Consider the following two statements GIT2011MCQ18
A - Twisted Pair Cables are more expensive than Fibre Optic cables
B - Fibre Optic cables transmit data at a high speed than Twisted Pair Cables
Of the above statements
1)Both A and B are true 2)A is true and B is false
3)A is false and B is true 4)Both A and B are false
47. Assume that you are planning to network the computer laboratory of your school. What is the most appropriate type of network for this purpose? GIT2011MCQ19
1)Wide Area Network (WAN) 2)Metropolitan Area Network (MAN)
3)Local Area Network (LAN) 4)all of the above

48. Which of the following is an advantage/are advantages of computer networks?
GIT2011MCQ20
A - Sharing data
B - Sharing of resources and printers
C - Improving the security of data storage
1)A only 2)B only 3)A and B only 4)all A, B and C
49. A person playing computer games using a new computer complains that the movements in the game are slow. GIT2011MCQ40
Which of the following should be installed to overcome the problem?
1)A sound card with better specification 2)A hard disk with larger capacity
3)A dedicated graphics card with better specification 4)A network Interface Card
50. Answer Sheet of Multiple choice questions paper can be read with AL 2011MCQ03
1)Digitizer 2)Light Pen 3)MICR 4)Scanner 5)POS terminal
51. The abbreviation GSM stands for AL 2011MCQ06
1)Global Systems for Mobile Access 2)Global System for Mobile Communication 3)Global System for Mobile Phone 4)Global System for Mobile Transmission 5)Global System for Mobile Interconnection
52. The main role of a "router" in a computer network is AL2011MCQ08
1)To broadcast data packets to all hosts in the network
2)To monitor the network for malicious activities
3)To retransmit the received signal at a higher power
4)To allow hosts in two networks to communicate
5)To allow load balancing between multiple separate networks
53. Consider the following memory types AL2011MCQ30
A -Read only memory
B -Secondary memory
C -Cache memory
D -Flash memory
E -Random Access Memory
Which of the above can be considered a Volatile Memories?
1)A and B only 2)A and C only 3)C and D only 4)C and E only 5)D and E only
54. A student prepares a report using word-processing software. What is the most suitable device that can be used to insert a picture book into the soft copy of the report? OL2012MCQ04
1)Keyboard 2)Mouse 3)Printer 4)Scanner
55. Which of the following represents an input device, a storage device and an output device respectively? GIT2012MCQ05
1)Keyboard, Hard disk, Microphone 2)Mouse, Keyboard, Printer
3)Keyboard, Touch Pad, Mouse 4)Scanner, Flash Memory, Speaker
56. Amal has connected his mobile phone to a notebook computer without using physical cables to transfer some photographs from his mobile phone to the notebook computer. Which of the following types of communication media is/are used for this purpose? GIT2012MCQ09
A - Guided Media B - Unguided Media C - Bounded Media D - Unbounded Media
1)A and B only 2)A and C only 3)B and C only 4)B and D only
57. School Net is a network that connects 1000 secondary schools, 100 computer resource centres and many other institutes relating to education across Sri Lanka. Which of the following network types is most appropriate for School Net? GIT2012MCQ10
1)Local Area Network 2)Metropolitan Area Network
3)Wide Area Network 4)Bluetooth network
58. A network cable needs to be laid in an area where Electro Magnetic Interference (EMI) is present. Which of the following cable types is most suitable for this purpose? GIT2012MCQ11
1)Shielded Twisted Pair (STP) 2)Unshielded Twisted Pair (UTP)
3)Coaxial 4)Fiber Optic
59. "The data in is read by using the Laser technology" AL2012MCQ03
Which of the following is most appropriate to fill the blank in the above statement?
1) Floppy Disk 2)Magnetic Tape 3)Compact Disk 4)Magnetic Hard Disk 5)Flash Memory

60. The component that decodes the instructions fetched into the CPU is called the AL2012MCQ28
1)Primary Memory 2)Register Unit 3)Control Unit 4)ALU 5)Program Counter
61. When a program is executed, instructions and data are directly bought into the Central Processing Unit through the cache memory from the OL2013MCQ06
1)CD 2)Hard Disk 3)Main memory 4)Computer Network
62. Which of the following are correct regarding computer networks? OL2013MCQ34
A - Accessing shared resources may be difficult when the network is broken or congested.
B - Two or more computers must be **always** connected with a cable in creating a computer network.
C - Software can be centrally managed in a computer network.
1)A and B only 2)A and C only 3)B and C only 4)All A, B and C
63. Which of the following statements are correct regarding the data transmission modes? OL2013MCQ35
A - Full duplex mode allows transmission of data both directions at the same time.
B - Half duplex mode allows transmission of data in both directions at the same time.
C - Simplex mode allows transmission of data only in one direction.
1)A only 2)B only 3)A and C only 4)B and C only
64. The Central Processing Unit (CPU) consists of GIT2013MCQ02
1)Arithmetic Unit and Control Unit 2)Logic Unit and Control Unit
3)Arithmetic and Logic Unit (ALU) and Control Unit 4)Arithmetic and Logic Unit

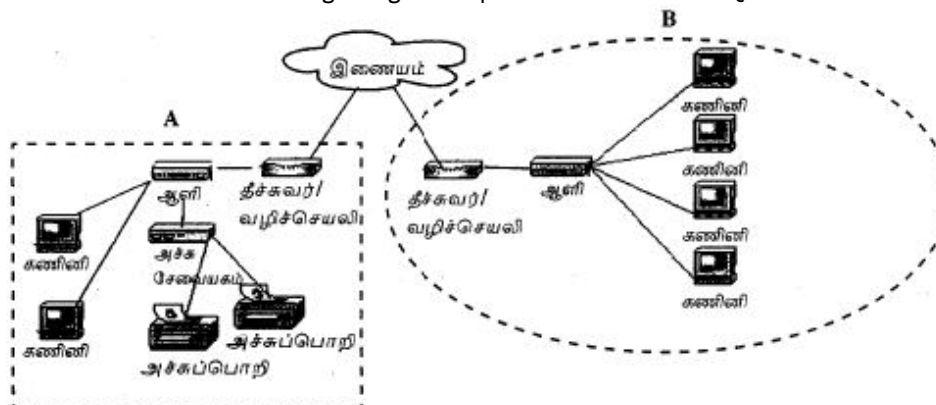
65. Consider the following images A, B and C showing different types of network cables: GIT2013MCQ07



Which of the following correctly represents A, B and C?

- 1)A=Coaxial, B=fibre optic, C=Twisted Pair 2)A=Fibre optic, B=Coaxial, C=twisted pair
3)A=Coaxial, B=twisted pair, C=fibre optic 4)A=Fibre optic, B=twisted pair, C=coaxial
66. Which of the following can be considered as examples of data communication? GIT2013MCQ08
A - Communication between two computers over the Local Area Network (LAN)
B - Communication over the Internet
C - Sending a text message (SMS) using mobile phone
1)A and B only 2)B and C only 3)A and C only 4)All A, B and C
67. Microwave, Infrared wave and Radio wave are examples of GIT2013MCQ09
Which of the following terms is suitable to fill the blanks in the above statement?
1)guided media 2)bounded media 3)unguided media 4)unprotected media
68. Which of the following statement(s) is/are **true** about digital signal? GIT2013MCQ12
A - Human voice in face to face communication is an example for a digital signal
B - Digital signals are represented by square waves
C - A sea wave is an example for a digital signal
1)A only 2)B only 3)C only 4)B and C only
69. Which of the following are the factors to be considered in selecting a data transmission media for a network? GIT 2013MCQ13
A - data transmission rate B - Distance C - Cost and ease of installation
1)A and B only 2)B and C only 3)C only 4)All A, B and C

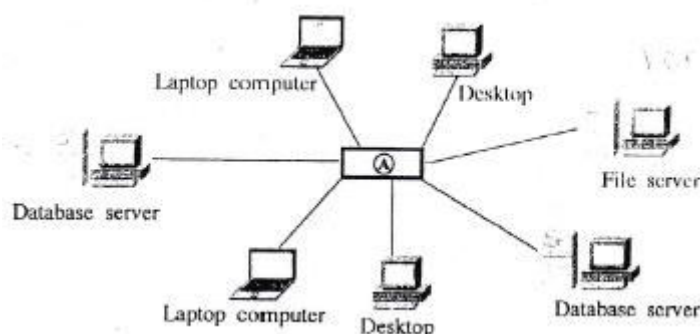
70. A and B in the following diagram shows networks in two countries. What type of network does the following diagram represents? GIT2013MCQ14



- 1)Local Area Network 2)Metropolitan Area Network 3)Wide Area Network 4)Mobile Network
71. A high speed temporary storage, which is part of the microprocessor that holds data and instructions during the execution, is called AL2013MCQ03
1)Registers 2)RAM 3)Virtual Memory 4)EPROM 5)Flash memory
72. Microprocessors are usually compared by their clock speed, measured in or by their word size, measured in That can be processed in a single clock cycle. Which of the following is most appropriate to fill the blanks in the above statement? AL2013MCQ04
1)Bits, Megahertz 2)Bytes, Gigahertz 3)Gigahertz, Bytes 4)Megahertz, Bits 5)Seconds, Bits
73. Typically the cache memory is used to store AL2013MCQ05
1)a large volume of data temporarily
2)the least frequently accessed data permanently
3)the least frequently accessed data temporarily
4)the most frequently accessed data temporarily
5)the most frequently accessed data permanently
74. Which of the following components is located outside the microprocessor? AL2013MCQ08
1)Arithmetic Logic Unit (ALU) 2)RAM 3)Control Unit 4)Registers 5)Level 1 cache memory
75. Consider the following devices: OL2014MCQ02
A - Digital camera, B - Speaker, C - Hard disk
Which of the above is/are usually used as an output device/devices of a computer?
1) A only 2)B only 3)A and B only 4)B and C only
76. Which of the following is correct during the program execution on the CPU? OL2014MCQ03
1)Instructions and data are brought from the main memory.
2)Instructions are brought from the main memory and the data are brought from the secondary storage.
3)Instructions are brought from the secondary storage and data are brought from the main memory.
4)Instructions and data are brought from the secondary storage.
77. Kasun who lives in Kandy has created a digital art as an image file on his computer. He wants to share it with his four friends living in Colombo, Jaffna, India and Japan. Which of the following method/s is/are suitable for Kasun to share the image file with his friends? OL2014MCQ04
A - using file transfer software over the LAN to send the image file with his friends' computers
B - using email
C - copying the file to four USB flash drives and sending them to his friends.
1)A only 2)B only 3)C only 4)A and C only

78. Which of the following are components of the Central Processing Unit (CPU) of a computer? GIT2014MCQ02
 1)Arithmetic and Logic Unit (ALU), Control Unit
 2)Arithmetic and Logic Unit, Secondary Storage
 3)Control Unit, Main Memory
 4)Main Memory, Secondary Storage
79. Which of the following are examples of secondary storage? GIT2014MCQ03
 1) CD, Flash memory, Magnetic Tape
 2) CD, Magnetic Tape, Read Only Memory (ROM)
 3) DVD, Magnetic Tape, Random Access Memory (RAM)
 4) Flash Memory, Random Access Memory, Read Only Memory
80. Which of the following media provides the highest speed of communication? GIT2014MCQ27
 1)Co-axial cable 2)fibre optic cable 3)radio-wave 4)twisted pair cable
81. Which of the following media provides the highest level of mobility for a user of modern laptop within a LAN? GIT2014MCQ28
 1)co-axial cable 2)fibre optic cable 3)radio wave 4)twisted pair cable
82. Which of the following is **not** useful for a desktop computer in connecting to the internet? GIT2014MCQ29
 1)email software 2)Internet Service Provider 3)Network Interface Card 4)router
83. Which of the following cannot be used to connect few laptops together in a LAN? GIT2014MCQ30
 1)external disk drive 2)hub 3)router 4)switch
84. Which of the following converts digital data to analog data to transmit over an analog telephone network? AL2014MCQ07
 1) Network Interface Card (NIC) 2)Modem 3)Multiplexer 4)Bluetooth adaptor 5)Wi-Fi card
85. The Central Processing Unit (CPU) of a computer consists of which of the following? OL2015MCQ01
 1)Arithmetic and Logic Unit (ALU), Control Unit (CU), display unit
 2)Arithmetic and Logic Unit (ALU), Control Unit (CU), input/output devices
 3)Arithmetic and Logic Unit (ALU), Control Unit (CU), memory registers
 4)Arithmetic and Logic Unit(ALU), Control Unit (CU), secondary storage
86. Which of the following is not a potential benefit of a computer network which is connected to the Internet? OL 2015MCQ02
 1)Backing up files to a file server in another network connected to the internet
 2)reduced risk of viruses and malware
 3)Sharing resources in the network
 4)Use of the web
87. In which of the following situation/s will a tablet computer be suitable? GIT2015MCQ01
 A - for a sales representative to use while he is travelling
 B - to be used in a school office to enter daily attendances data of students
 C - to enter data during a field visit
 1) A only 2)B only 3)A and B only 4)A and C only
88. Consider the following statements GIT2015MCQ08
 A - infrared transmission is an example for the use of unguided media
 B - Microwave transmission generally used for long distance data communication is an example for the use of radiated data.
 C - Fibre optic cables are examples for unguided media.
 Which of the above statement/s is/are true?
 1)A only 2)B only 3)A and B only 4)B and C only
89. Which of the following media uses light waves for communication? GIT2015MCQ30
 1)co-axial cable 2)Optical Fiber 3)Twisted pair 4)USB cable
90. Consider the following media GIT2015MCQ31
 A - Twisted pair cable B - USB cable C - optical fiber
 Which of the above media are suitable to be used to connect a laptop computer directly to a standalone printer?
 1)A and B only 2)A and C only 3)A and B only 4)All A, B and C

91. Which of the following are advantages of a computer network compared to a standalone desktop? GIT2015MCQ32
 A - sharing of data among machine on the network
 B - reduced risk of malware attacks on the machines connected to the network
 C - sharing of software application on the central server
 1) A only 2) B only 3) A and B only 4) A and C only
92. Random Access Memory (RAM) modules are often compared by their capacity, measured in and by their speed, measure in most suitable words to fill the blanks of the above statements are respectively. AL 2015MCQ05
 1) Kilobytes, Gigabytes 2) Gigabytes, Megabits per second 3) Gigabytes, Megahertz
 4) Megahertz, Kilohertz 5) Gigabits, Megabits per second
93. Consider the following devices OL2016MCQ02
 A - Mouse
 B - Touch Screen
 C - External DVD writable drive
 Which of the above can be used as both input and output devices?
 1) A and B only 2) A and C only 3) B and C only 4) All A, B and C
94. Consider the following statements OL2016MCQ30
 A - Optical Fibre cables provide faster data transfer rates than that of the copper cables
 B - Copper cables provide faster data transfer rates than that of optical fibre cables
 C - Copper cables are more expensive than Optical Fibre cables Which of the above is/are correct?
 1) A only 2) B only 3) A and C only 4) B and C only
95. Consider the following diagram GIT2016MCQ03
 Which of the following device represents(A) in the local area network as shown in the above diagram.



- 1) Gateway 2) Network Interface Card 3) Switch 4) Modem
96. Consider the following statements GIT2016MCQ12
 A - Twisted Pair cable use copper as the medium to transmit data
 B - Optical fiber cables use light waves to transmit data
 C - Data transmission speed of twisted pair cable is higher than the optical fibre cable
 Which of the following is correct?
 1) A and B only 2) A and C only 3) B and C only 4) All A, B, C
97. Which of the following component of a Central Processing Unit (CPU)? AL2016MCQ02
 1) ROM 2) RAM 3) ALU 4) L3 Cache memory 5) Power Supply Unit
98. The content stored in is read by using the optical technology. AL2016MCQ05
 Which of the following answers is the most appropriate to fill the blanks in the above statement?
 1) flash memory 2) floppy disk 3) magnetic tape 4) compact disk 5) hard disk
99. In modern computers, multiple levels of cache memory is used to optimize the performance. Among them, the Cache which is on/in Is the fastest and most expensive cache memory. AL2016MCQ06
 Which of the following terms are correct to fill the blanks respectively in the above statement?
 1) Level 1 (L1), main memory 2) Level 3 (L3), motherboard 3) Level 1 (L1), microprocessor
 4) Level 2 (L2), microprocessor 5) Level 3 (L3), microprocessor

100. Which of the following statements is correct with respect to a scanner? AL2016MCQ35
 1)A scanner is a software which scans a printed document and converts them into digital images.
 2)A scanner is an output device of a computer
 3)The optical character reader (OCR) software is an essential component of a scanner
 4)A scanner is an input device of a computer
 5)Scanners are used to store moving pictures in digital form
101. Which of the following communication media uses the reflection of light to transmit data? OL2017MCQ05
 1)Unshielded Twisted Pair (UTP) 2)Optical Fiber3)Coaxial Cable 4)Shielded Twisted Pair (STP)
102. Some devices serve as either input or output devices only while others serve as both input and output devices. Which of the following serve as output device only? OL2017MCQ06
 1)Touch screen display and scanner 2)Printer and Webcam
 3)Printer and Speaker 4)Touch screen display and barcode reader
103. Consider the following statements regarding the operations of a computer: GIT2017MCQ01
 A - The main memory loses its data when the computer is turned off
 B - The hard disk retains its data already stored in it. When the computer is turned off
 C - Instructions are brought from secondary storage to main memory for processing.
 Which of the above statements are true?
 1)A only 2)A and C only 3)B and C only 4)All A, B and C
104. Which of the following device is used to connect a Local Area Network (LAN) to a Wide Area Networks (WAN)? GIT2017MCQ02
 1)network interface card 2)router 3)server 4)hub
105. Which of the following cables is most suitable to connect a desktop to a hub? GIT2017MCQ03
 1)coaxial cable 2)telephone cable 3)UTP cable 4)USB cable
106. If you are planning to network classrooms and laboratories of your school, what is the most suitable type of network that should be implemented? GIT2017MCQ04
 1)LAN 2)MAN 3)VPN 4)WAN
107. Which of the following are benefits of a computer network over standalone computer? GIT2017MCQ09
 A - Sharing a hardware device such as a printer attached to the network
 B - Reduce risk of virus attacks on computers connected to the network
 C - Sharing of files and directories among machines on the network
 1)A and B only 2)A and C only 3)B and C only 4)All A, B and C
108. Which of the following lists provides **only** technical specifications of a laptop computer? GIT2017MCQ31
 1)number of USB ports, after sales service, capacity of RAM, size of the display
 2)processor speed, hard disk capacity, price, capacity of RAM
 3)processor speed, size of the display, number of USB ports, hard disk capacity
 4)warranty period, reputation of the seller, number of USB ports, weight of the laptop.
109. Who proposed the stored program concept first? AL2017MCQ01
 1)Lady Ada Augusta 2)Charles Babbage 3)Howard Aiken 4)Blaise Pascal 5)Von Neumann
110. Which of the following components is generally seen outside the Central Processing Unit (CPU) of a computer? AL2017MCQ02
 1)RAM 2)Control Unit 3)ALU 4)General Purpose Registers 5)L1 cache memory
111. A On motherboard is used to expand the functionality of a computer. Which of the followings is the most appropriate to fill in the blank in the above statement? AL 2017MCQ12
 1)Bus 2)Clock 3)RAM 4)Slot 5)ROM

112. Consider the following data input devices AL2017MCQ39
 A - On screen keyboard (virtual keyboard)
 B - Bar code reader
 C - Magnetic card reader
 Which of the above devices can be used to input data more efficiently?
 1)A only 2)B only 3)C only 4)A and B only 5)B and C only
113. Which of the following is suitable for durable storage of data in a computer? OL2018MCQ19
 1)cache memory 2)hard disk 3)RAM 4)video memory
114. Which of the following lists **only** input devices? OL2018MCQ20
 1)Keyboard, barcode reader, CD ROM, flatbed scanner
 2)keyboard, webcam, touch screen, flatbed scanner
 3)mouse, keyboard, touch screen, flatbed scanner
 4)mouse, webcam, touch screen, flatbed scanner
115. Which of the following can be used to measure the clock speed of the Central Processing Unit of a computer? OL2018MCQ21
 1)Gigabyte (GB) 2)Gigahertz(GHz) 3)Megabits per second (Mbps) 4)Revolutions per minute (RPM)
116. Consider the following statements. AL2018MCQ04
 A - Word size is the number of bits processed by the CPU of a computer in a single action (instance).
 B - Data bus width and register width are directly related to word size of a computer.
 C - Word size of modern general purpose computer is either 32 or 64 bits.
 Which of the above statements is/are correct?
 1)A only 2)B only 3)C only 4)B and C only 5)All A, B and C
117. Consider the following computer memory types. AL2018MCQ13
 A - CMOS memory
 B - cache memory
 C - flash memory
 D - hard disk
 E - RAM
 F - registers
 Which of the above are volatile memory types?
 1)A, C and D only 2)A, D and E only 3)A, E and F only 4)B, E and F only 5)C, E and F only
118. Which of the following devices have both input and output capabilities? OL2019MCQ01
 1)Joystick 2)OCR device 3)touch screen 4)webcam
119. Which of the following can be good practice to protect data and information from computer hard disk failures? OL2019MCQ02
 1)installing a firewall 2)installing an antivirus software
 3)taking periodic backup 4)using strong password
120. Select the most suitable computer type from *super computer, a Laptop computer and a tablet computer*, for the following applications: OL2019MCQ06
 A - to process very large amounts of data that are continuously obtained through satellites
 B - for a writer to work on essay during a vacation away from home
 C - for a travelling sales representative to enter item requests while visiting shops
 1)A:laptop computer, B:Super Computer, C:tablet computer
 2)A:Laptop computer, B:tablet computer, C:super Computer
 3)A:Super computer, B:Laptop computer, C: tablet computer
 4)A:tablet computer, B::Laptop computer, C:laptop computer
121. Consider the following statement with blanks labelled A and B
 When processing, the CPU uses its A to temporarily store data that are brought from B
 Which of the following combinations is suitable to fill in the blanks labeled A and B respectively OL2019MCQ07
 1)primary memory, registers 2)registers, primary memory
 3)secondary memory, primary memory 4)secondary memory, registers
122. Which of the following are correct regarding transmission media? OL2019MCQ08
 A - Unshielded Twisted Pair (UTP) cables are suitable to transmit data for long

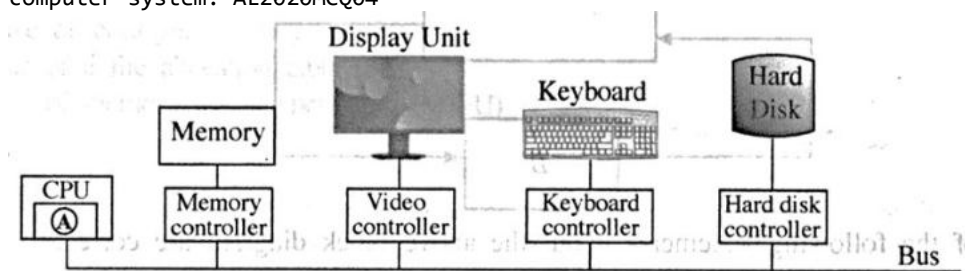
distances over 200m

B - Fiber optic cables transmit data faster than UTP Cable

C - Infrared data transmission is used in wireless keyboards to communicate with computers

1)B only 2)C only 3)B and C only 4)all A, B and C

123. In a particular network, each node is connected directly to a central network device. This topology is referred to as a AL2019MCQ04
1)bus 2)hybrid 3)mesh 4)ring 5)star
124. Which of the following options contains only guided media? AL2019MCQ26
1)Coaxial, Fiber optics and Infrared
2)Coaxial, Fiber optics and Microwave
3)Coaxial, Fiber optics and Twisted pair
4)Coaxial, Infrared and Twisted pair
5)Fiber optics, Satellite communication and Twisted pair
125. Which of the following contains only input device? OL2020MCQ01`
1)Multimedia projector, Printer, Keyboard, Mouse
2)Printer, Keyboard, Touch Screen, Joystick
3)Monitor, Light Pen, Multimedia projector, Keyboard
4)Mouse, Keyboard, Light pen, Joystick
126. Which of the following are true about the secondary memory of a computer system? OL2020MCQ05
A - Data is not erased if the computer is switched off
B - Solid state devices can be used as secondary memory
C - Secondary memory is a part of the CPU memory
1)A and B only 2)A and C only 3)B and C only 4)All A, B and C
127. In which of the following network topologies, each computer is exactly connected with two other computers? OL2020MCQ30
1)Bus 2)Ring 3)Mesh 4)Star
128. A computer processor will operate fastest when the data that it wants is in the AL2020MCQ01
(1) cache memory. (2) hard disk. (3) magnetic tape.
(4) main memory. (5) optical disk.
129. Which of the following hardware components will lose data when the power to a computer is switched off? AL2020MCQ02
A - registers
B - cache memory
C - main memory
(I) A only (2) A and B only (3) A and C only (4) B and C only (5) All A, B and C
130. Consider the following diagram showing some hardware component connections on a computer system: AL2020MCQ04



The hardware part within the CPU indicated by ① in the above diagram has a set of registers that has the memory translation maps of the currently maps of the currently running process. When given an input virtual address of the current process it outputs the relevant physical address (if any)

The ① in the above diagram denotes the

- 1)arithmetic and logic unit (ALU) 2)control unit 3)L1 cache memory
4)memory management unit 5)page table.

Answers

- Hardware (Software is a collection of instructions and data that tell a computer how to work. Firmware used to start the computer like BIOS. Live ware is another name for Computer Users)
- CPU (Hard disk is one of the secondary storage devices used to store also known non-volatile memory as it keeps the data even after the computer shutdown or power

interruptions, does not involved with processing of data, RAM is volatile memory used to keep the data temporarily does not process data. Data bus used to transfer data which you can imagine like bus transports the people here data bus transport data).

3. All A, B and C
4. Microcomputer (You can guess the answer as the term Microsoft coined from **Micro**computer **Software** as laptop used to have Microsoft software).
5. GHz (gigahertz)
6. It is the place where arithmetic and logic operations are done.
7. LAN
8. Star
9. All A, B, and C
10. Joystick (Joystick is a device 🖱 not a port)
11. Laser printer is an output device
12. PDA (Due to the advancements of smartphones you cannot see PDA these days).
13. Barcode reader
14. Simplex
15. Bus
16. Computer Aided Design
17. Firewall
18. Incorrect on is PS2 (Can only connect Keyboard and Mouse), Parallel Port used to connect old printers and USB used to connect new printers.
19. Web Brower
20. Software for email communication
21. B (Harddisk - Storage, Scanner- Input)
22. (A) and (C)
23. half duplex communication of data through a data channel
24. Incorrect one is "Consist of many shared data communication medium with various interfaces"
25. data server (Database servers also known as data servers)
26. Laser Printer
27. Printer
28. False one is "A mouse is an input device that translate its movement on the desktop into analog form". Though the movement of the mouse could be analogue but they send digital signal to the computer.
29. A webserver
30. Input Devices A and B only (A - Keyboard B - Scanner C - Printer D - Speaker)
31. A and B only (A - Random Access Memory (RAM) is a volatile memory.
B - Hard disk is an auxiliary storage device.
C - A floppy disk can store more data than a compact disk (CD)
32. A = power socket, B= PS/2 port, C= USB port
33. Arithmetic and Logic Unit (ALU) and Control Unit (CU).
34. Desktop Computer
35. b only (Answer needs to be checked)
36. Network Interface Card
37. All a, b, c can connect to a network (a - Printer b - Scanner c - Server)
38. a only (a - Fibre optics cables provide faster transfer rates than copper cables.
b - Copper cables provide faster transfer rates than fibre optic cables.
c - Copper cables are expensive than the fibre optics cables.)
39. Installing firewall
40. Hard disk and Compact Disk (CD)
41. Input/Output device only
42. Keyboard, Speaker, Compact Disk Drive
43. Both B and C ((A) - Desktop Computer (B) - Palmtop Computer (C) - Laptop Computer)
44. Control Unit and Arithmetic and Logic Unit (ALU)
45. Switch, Router
46. A is false and B is true (A - Twisted Pair Cables are more expensive than Fibre Optic cables,
B - Fibre Optic cables transmit data at a high speed than Twisted Pair Cables)
47. Local Area Network (LAN)
48. A and B only
49. A dedicated graphics card with better specification (Sound cards will not affect the movement of in games)
50. MICR
51. Global System for Mobile Communication
52. to allow hosts in two networks to communicate
53. C and E only are considered as volatile memory (A -Read only memory, B -Secondary memory, C -Cache memory, D -Flash memory, E -Random Access Memory).
54. Scanner
55. Scanner, Flash Memory, Speaker (considered as input, storage and output device respectively)

56. B and D only
57. Metropolitan Area Network
58. Fibre Optic
59. Compact Disk
60. Control Unit
61. Main memory
62. A and C only (Statement B is incorrect as we can connect two computers via Wi-Fi or Bluetooth without cable).
63. A and C only
64. Arithmetic and Logic Unit (ALU) and Control Unit
65. A=Coaxial, B=fibre optic, C=Twisted Pair
66. All A, B and C
67. unguided media
68. B only
69. All A, B and C
70. Wide Area Network
71. Registers
72. Megahertz, Bits
73. the most frequently accessed data temporarily
74. RAM
75. B only
76. Instructions and data are bought from the main memory.
77. B only (Statement A is incomplete as we cannot come to a conclusion that it is connected to the internet or not. So we have to choose the most suitable ones).
78. Arithmetic and Logic Unit (ALU), Control Unit
79. CD, Flash memory, Magnetic Tape
80. fibre optic cable
81. radio wave
82. email software
83. external disk drive
84. Modem
85. Arithmetic and Logic Unit (ALU), Control Unit (CU), memory registers
86. reduced risk of viruses and malware
87. A and C only
88. A and B only
89. Optical Fiber
90. A and B only
91. A and C only
92. Gigabytes, Megahertz
93. B and C only
94. A only
95. Switch
96. A and B only
97. ALU
98. compact disk
99. Level 1 (L1), microprocessor
100. A scanner is an input device of a computer
101. Optical Fiber
102. Printer and Speaker
103. All A, B and C
104. router
105. UTP cable
106. LAN
107. A and C only
108. processor speed, size of the display, number of USB ports, hard disk capacity
109. Von Neumann
110. RAM
111. Slot
112. B and C only
113. hard disk
114. Keyboard, barcode reader, CD ROM, flatbed scanner
115. Gigahertz(GHz)
116. All A, B and C
117. B, E and F only
118. touch screen
119. taking periodic backup
120. A:Super computer, B:Laptop computer, C: tablet computer
121. registers, primary memory
122. B and C only
123. Star
124. Coaxial, Fiber optics and Twisted pair
125. Mouse, Keyboard, Light pen, Joystick

- 126. A and B only
- 127. Ring
- 128. cache memory.
- 129. All A, B and C
- 130. memory management unit